

The empirical recurrence for OEIS A300212: recovering a conjecture that is not written down

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Abstract

OEIS A300212 records “Empirical recurrence of order 89”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1024$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 89, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 89 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A300212 is “Number of $n \times 5$ 0..1 arrays with every element equal to 0, 1, 2, 3, 4, 5, 7 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 512, 11965, 298600, 7452385, 186427449, 4666136435, 116802113451, ...

The entry states:

Empirical recurrence of order 89 (see link above).

As of the “Last modified” line on the live entry (Dec 27 2023 21:09:46, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1024$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2048$ exact terms of the model returns order 89 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{89} c_i a(n-i),$$

c_1	28	c_{24}	219186730623284	c_{47}	1856199584570583	c_{70}	−112498395158512200
c_2	−45	c_{25}	279714431374639	c_{48}	15599535129557728	c_{71}	−87786579733391139
c_3	−539	c_{26}	−456943921700101	c_{49}	−58990789953403560	c_{72}	−61830687238141749
c_4	−5727	c_{27}	−2037988131019345	c_{50}	−213637267629605718	c_{73}	−42597175062109567
c_5	17612	c_{28}	−2144062503860278	c_{51}	−326724158151554380	c_{74}	−30626563990568297
c_6	107931	c_{29}	2881665746954329	c_{52}	−190289423965888208	c_{75}	−22688961809118377
c_7	98450	c_{30}	10871966028596660	c_{53}	245085644636661070	c_{76}	−13632604370769026
c_8	−1233830	c_{31}	9571915503884741	c_{54}	606796817198249956	c_{77}	−5544943544294921
c_9	97107	c_{32}	−10188045423486889	c_{55}	514487267440408529	c_{78}	−955955997162154
c_{10}	19699825	c_{33}	−32431780266021529	c_{56}	−91900330673284523	c_{79}	547885666733060
c_{11}	−25571306	c_{34}	−24359136369854826	c_{57}	−816914800972198759	c_{80}	486452182452806
c_{12}	−422400144	c_{35}	15143157788069771	c_{58}	−974432394219237855	c_{81}	171126295603554
c_{13}	−825313677	c_{36}	44049150186563090	c_{59}	−362350125674436035	c_{82}	21066338746732
c_{14}	3096169879	c_{37}	36985968064372038	c_{60}	442932390954453991	c_{83}	−7780505205944
c_{15}	10721092002	c_{38}	23854115225444654	c_{61}	803281588286135066	c_{84}	−3612283597668
c_{16}	−4441090770	c_{39}	11629508067902933	c_{62}	619708665743035237	c_{85}	−376217041156
c_{17}	−37150843169	c_{40}	−46974844098783720	c_{63}	133842339674672917	c_{86}	66370731696
c_{18}	207479768960	c_{41}	−134568745073659908	c_{64}	−124770235729814417	c_{87}	18368847424
c_{19}	847356185047	c_{42}	−117069966571638622	c_{65}	−24853844986192025	c_{88}	1309845504
c_{20}	−1246443458558	c_{43}	49643135720133134	c_{66}	82736831691617119	c_{89}	21659136
c_{21}	−12350649455577	c_{44}	179029307719103428	c_{67}	30246151187339881		
c_{22}	−21378455531454	c_{45}	142206020041161488	c_{68}	−53216719747644389		
c_{23}	39036807287787	c_{46}	31577742415503271	c_{69}	−103221773085248235		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 89, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $89 + 4$ terms removed returns order 89 as well, so $\deg q_{\min} = 89 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $89 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 89 that this sequence satisfies — and that family turns out to have exactly one member.

References

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